# MADHUBALA V

M.Sc. Applied Data Science | Analytics | AI / ML | Python | Gen AI | Tableau | SQL
(91)8838166131 | madhu69.vel@gmail.com | Chennai | LinkedIn | GitHub | Portfolio

## EDUCATION

**M.Sc. Applied Data Science** | SRM Institute of Science and Technology | **CGPA: 9.54** **2024 - 2026**
**B.Sc. Data Science** | Sri Ramachandra Faculty of Engineering and Technology | **CGPA: 8.05** **2020 – 2023**

## INTERNSHIP

**Data Science Intern** | L&T Construction **April 2022 – June 2022**

- Analyzed and visualized time-series data using Python to identify trends and patterns.
- Performed EDA and predictive modeling to generate meaningful data-driven insights.
- Applied preprocessing and feature engineering to improve data quality and modeling readiness.

## TECHNICAL SKILLS

**Languages:** Python | SQL | HTML | CSS
**Data Analysis:** EDA | Statistical Analysis | Predictive Analytics
**Visualization:** Tableau | Excel | Matplotlib | Seaborn
**Machine Learning:** Scikit-learn | TensorFlow | Keras
**Tools:** Jupyter Notebook | Google Colab
**Gen AI Tools:** ChatGPT | Google Gemini | DeepSeek | Canva

## CERTIFICATIONS

- **GenAI Powered Data Analytics Job Simulation** | Tata Group (Forage) | 2026
- **Deloitte Data Analytics Job Simulation** | Deloitte (Forage) | 2026
- **Statistics for Data Science |** Great Learning **|** 2024
- **SQL and Relational Database |** Cognitive Class (IBM) | 2023

## ACADEMIC PROJECTS

### Knee Osteoarthritis Detection

Python | TensorFlow | Keras | Transfer Learning

- Developed a 5-class osteoarthritis classifier using 8,260 X-ray images across KL Grades 0–4.
- Applied transfer learning, preprocessing, augmentation, and class balancing for robust classification.
- Evaluated model performance to improve prediction reliability across severity levels.

### Flipkart Sales & Customer Insights Dashboard

Tableau | Data Visualization | Business Analytics

- Analyzed 20K+ Flipkart records covering pricing, discounts, ratings, brands, and categories.
- Performed EDA to uncover product, pricing, and customer-rating trends.
- Built an interactive Tableau dashboard to present insights and support business decisions.

### AI Quote Generator & Image Caption Generator

Python | Gradio | Transformers | BLIP | gTTS

- Developed a Gradio application integrating category-based quote generation and image captioning.
- Implemented image-to-text captioning with text-to-speech functionality.
- Built interactive interfaces for both tools.